



**Università
di Genova**

High Performance Computing Report CUDA

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1 Architectures

For this project I have used two workstation, my personal one and the workstation provided by the University.

HP Envy 17-ae103nl (Laptop)

- **CPU:** Intel Core i7-6500U @ 2.50 GHz
- **Architecture:** x86_64
- **Core fisici:** 2
- **Thread logici:** 4 (Hyper-Threading)

University Workstation (Room 210)

- **CPU:** Intel Core i9-12900K (12th Gen)
- **Architecture:** x86_64
- **Core(s) per socket:** 16
- **CPU(s):** 24
- **Thread(s) per core:** 2
- **Socket(s):** 1
- **CPU max MHz:** 4.02 GHz
- **Profiling Tools:** Intel VTune Profiler, Intel Advisor

To use the Intel oneAPI tools, the environment was loaded with:

```
source /opt/intel/oneapi/setvars.sh
```

2 Hotspot Analysis

3 Vectorization

4 Parallelization

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